

Universidad Rafael Landívar

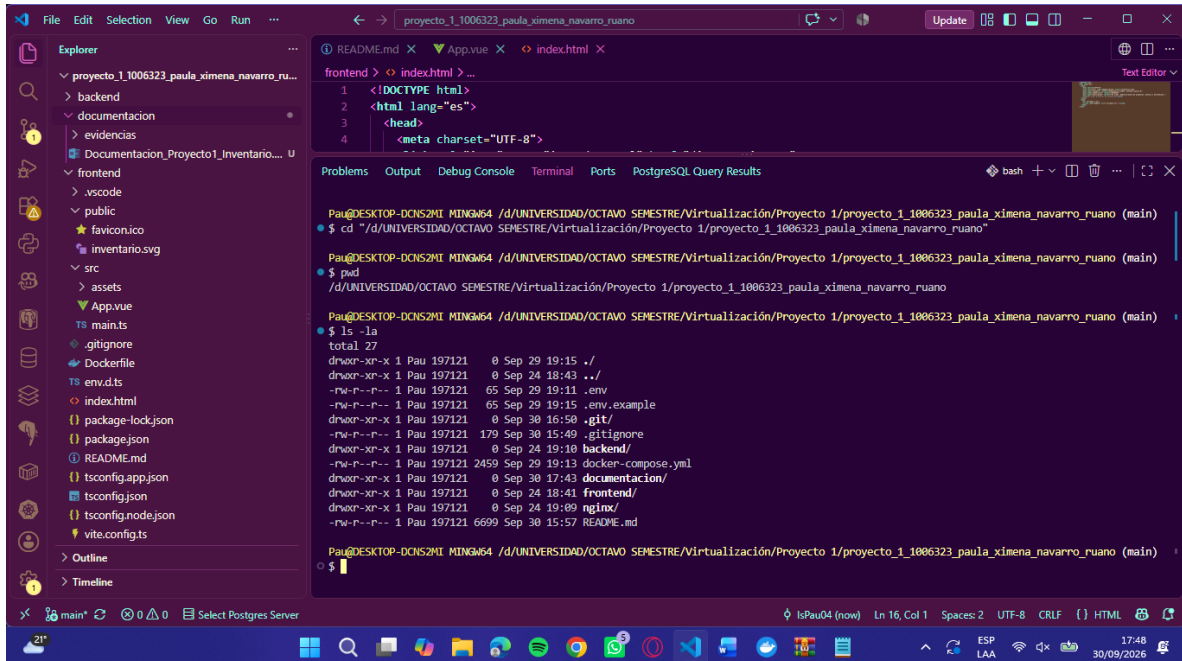
Facultad de Ingeniería

Ingeniería en informática y sistemas

Virtualización

Paula Ximena Navarro Ruano #1006323

Evidencias Capturas (Proyecto 1)



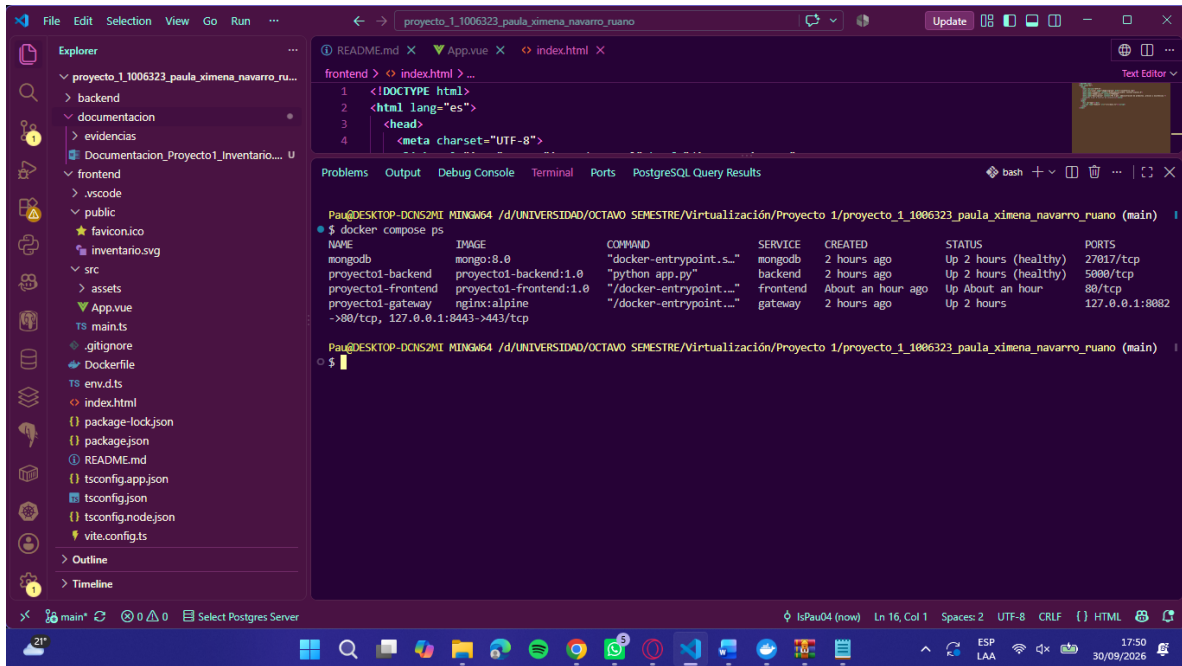
The screenshot shows a Visual Studio Code editor window with a project named 'proyecto_1_1006323_paula_ximena_navarro_ruano'. The Explorer sidebar on the left displays the project structure, including folders for 'backend', 'documentation', 'evidencias', 'frontend', and 'src', along with various configuration files like 'Dockerfile', 'env.d.ts', 'index.html', 'package-lock.json', 'package.json', 'README.md', 'tsconfig.app.json', 'tsconfig.json', 'tsconfig.node.json', and 'vite.config.ts'. The main editor area shows the 'index.html' file with the following content:

```
1 <!DOCTYPE html>
2 <html lang="es">
3 <head>
4 <meta charset="UTF-8">
```

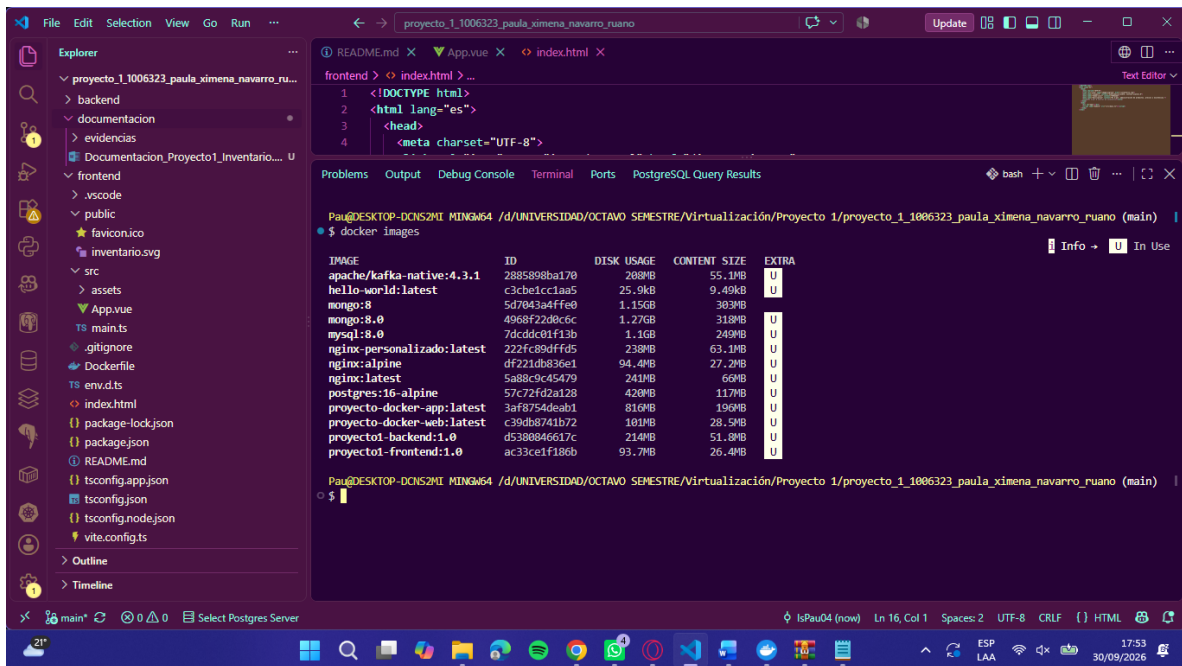
The terminal at the bottom shows the following commands and output:

```
Pau@DESKTOP-DCNS2MI MINGW64 /d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano (main)
$ cd "/d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano"
Pau@DESKTOP-DCNS2MI MINGW64 /d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano (main)
$ pad
/d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano
Pau@DESKTOP-DCNS2MI MINGW64 /d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano (main)
$ ls -la
total 27
drwxr-xr-x 1 Pau 197121  0 Sep 29 19:15 ./
drwxr-xr-x 1 Pau 197121  0 Sep 24 18:43 ../
-rw-r--r-- 1 Pau 197121  65 Sep 29 19:11 .env
-rw-r--r-- 1 Pau 197121  65 Sep 29 19:15 .env.example
drwxr-xr-x 1 Pau 197121  0 Sep 30 16:58 .git/
-rw-r--r-- 1 Pau 197121 179 Sep 30 15:40 .gitignore
drwxr-xr-x 1 Pau 197121  0 Sep 24 19:10 backend/
-rw-r--r-- 1 Pau 197121 2459 Sep 29 19:13 docker-compose.yml
drwxr-xr-x 1 Pau 197121  0 Sep 30 17:43 documentation/
drwxr-xr-x 1 Pau 197121  0 Sep 24 18:41 frontend/
drwxr-xr-x 1 Pau 197121  0 Sep 24 19:09 nginx/
-rw-r--r-- 1 Pau 197121 6699 Sep 30 15:57 README.md
Pau@DESKTOP-DCNS2MI MINGW64 /d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano (main)
$
```

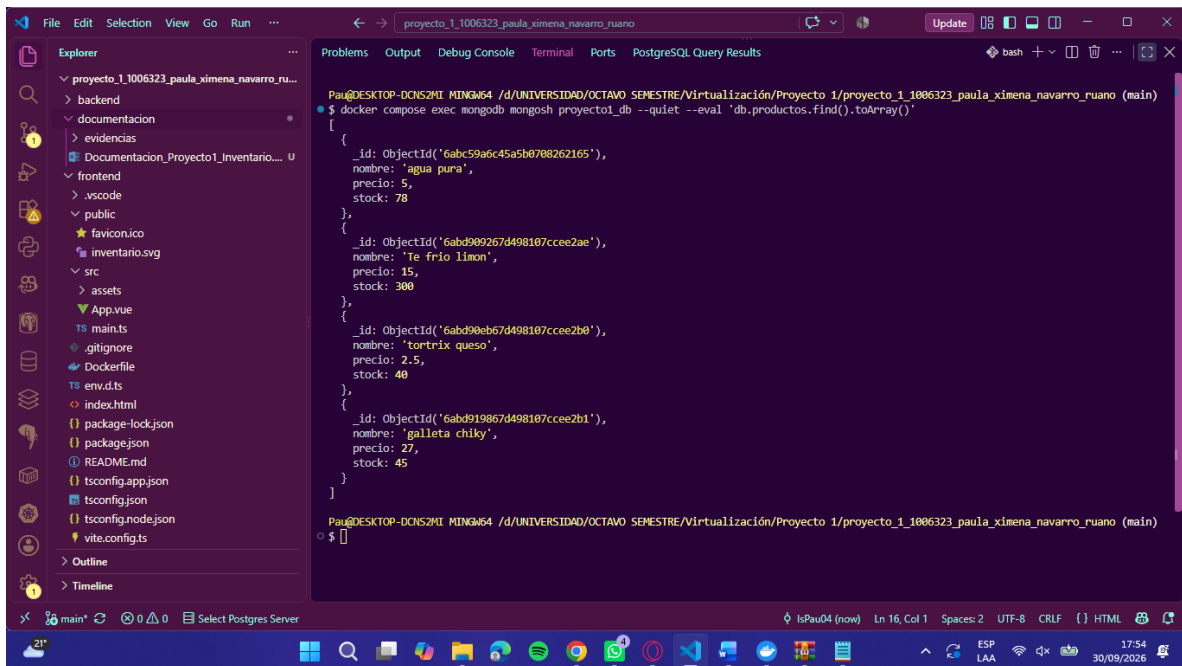
Verificación de la ubicación y estructura del proyecto mediante pwd y ls -la



Verificación del estado y los puertos de los servicios con Docker Compose



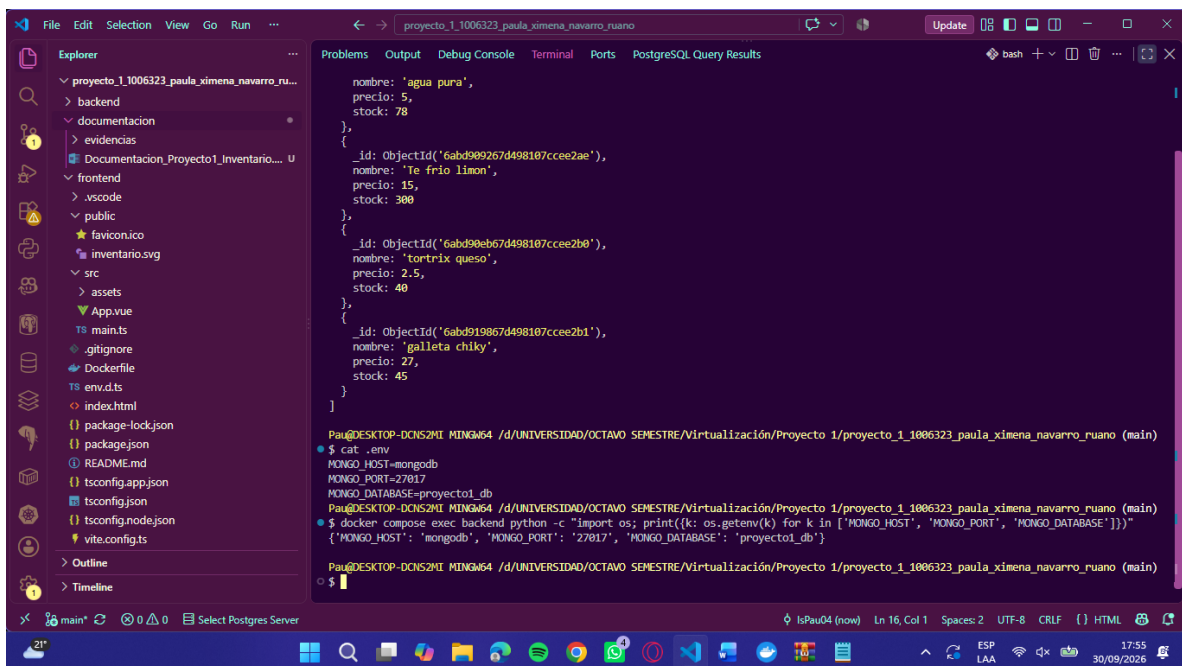
Verificación de las imágenes Docker del backend y frontend del proyecto.



The screenshot shows the Visual Studio Code interface with a project named 'proyecto_1_1006323_paula_ximena_navarro_ruano'. The Explorer sidebar on the left shows the project structure, including folders like 'backend', 'documentation', 'frontend', and 'src'. The main editor area displays the 'Terminal' tab, which shows the output of a MongoDB query executed via the 'docker compose exec mongodb mongosh' command. The query result is a JSON array of product objects.

```
Pau@DESKTOP-DCNS2HI MINGW64 /d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano (main)
$ docker compose exec mongodb mongosh proyecto1_db --quiet --eval 'db.productos.find().toArray()'
[
  {
    _id: ObjectId('6abc59a6c45a5b0708262165'),
    nombre: 'agua pura',
    precio: 5,
    stock: 78
  },
  {
    _id: ObjectId('6abd909267d498107ccee2ae'),
    nombre: 'Te frio limon',
    precio: 15,
    stock: 300
  },
  {
    _id: ObjectId('6abd90eb67d498107ccee2b0'),
    nombre: 'tortrix queso',
    precio: 2.5,
    stock: 40
  },
  {
    _id: ObjectId('6abd919867d498107ccee2b1'),
    nombre: 'galleta chiky',
    precio: 27,
    stock: 45
  }
]
```

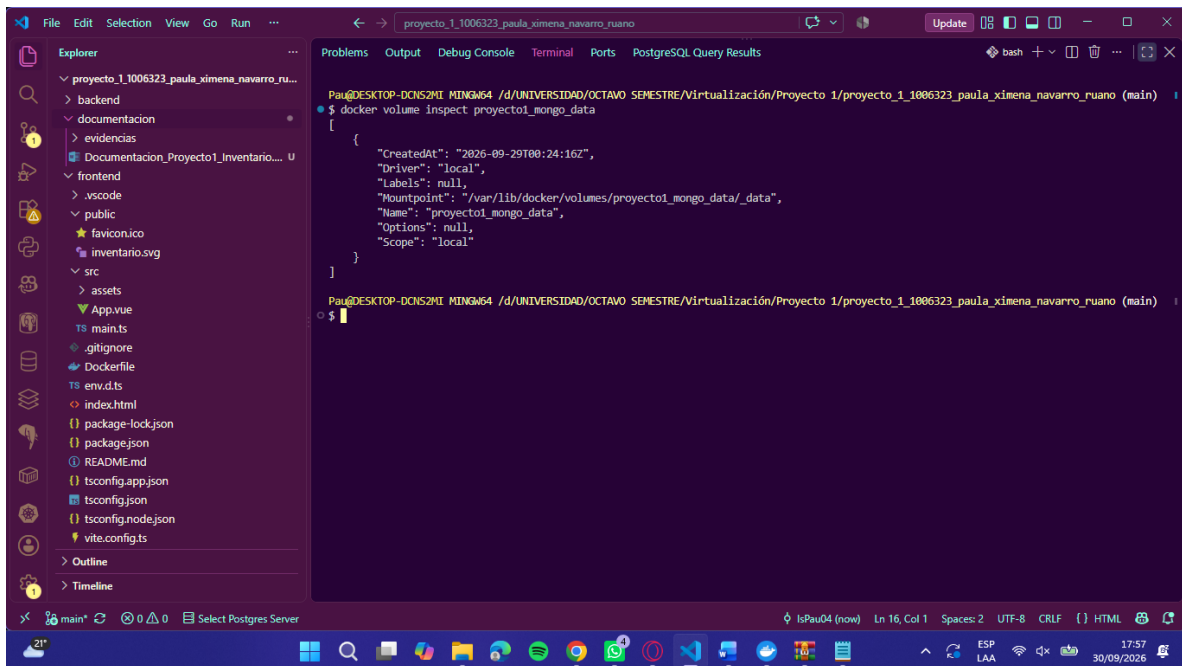
Consulta de los productos almacenados en MongoDB mediante mongosh.



The screenshot shows the Visual Studio Code interface with the same project. The 'Terminal' tab now displays the output of several commands used to verify the environment variables for the MongoDB connection. The commands include 'cat .env', 'docker compose exec backend python -c "import os; print((k: os.getenv(k) for k in ['MONGO_HOST', 'MONGO_PORT', 'MONGO_DATABASE']))"', and 'docker compose exec backend python -c "import os; print((k: os.getenv(k) for k in ['MONGO_HOST', 'MONGO_PORT', 'MONGO_DATABASE']))"'.

```
Pau@DESKTOP-DCNS2HI MINGW64 /d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano (main)
$ cat .env
MONGO_HOST=mongodb
MONGO_PORT=27017
MONGO_DATABASE=proyecto1_db
Pau@DESKTOP-DCNS2HI MINGW64 /d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano (main)
$ docker compose exec backend python -c "import os; print((k: os.getenv(k) for k in ['MONGO_HOST', 'MONGO_PORT', 'MONGO_DATABASE']))"
{'MONGO_HOST': 'mongodb', 'MONGO_PORT': '27017', 'MONGO_DATABASE': 'proyecto1_db'}
Pau@DESKTOP-DCNS2HI MINGW64 /d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano (main)
$
```

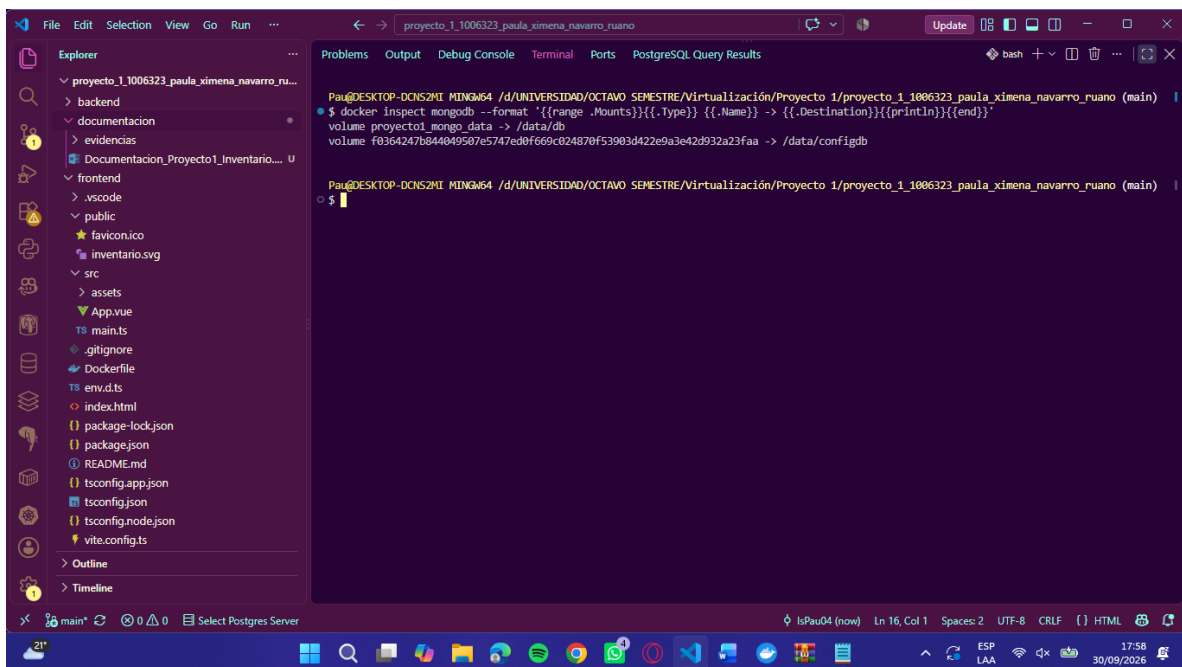
Verificación de las variables de entorno para la conexión del backend con MongoDB.



The screenshot shows a VS Code editor with a file explorer on the left and a terminal on the right. The file explorer shows a project structure for 'proyecto_1_1006323_paula_ximena_navarro_ruano'. The terminal shows the command 'docker volume inspect proyecto1_mongo_data' being executed, which returns a JSON object describing the volume's configuration, including its name, driver, and mountpoint.

```
Pau@DESKTOP-DCNS2HI MINGW64 /d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano (main)
$ docker volume inspect proyecto1_mongo_data
[
  {
    "CreatedAt": "2026-09-29T00:24:16Z",
    "Driver": "local",
    "Labels": null,
    "Mountpoint": "/var/lib/docker/volumes/proyecto1_mongo_data/_data",
    "Name": "proyecto1_mongo_data",
    "Options": null,
    "Scope": "local"
  }
]
```

Verificación del volumen Docker utilizado para almacenar los datos de MongoDB de forma persistente.

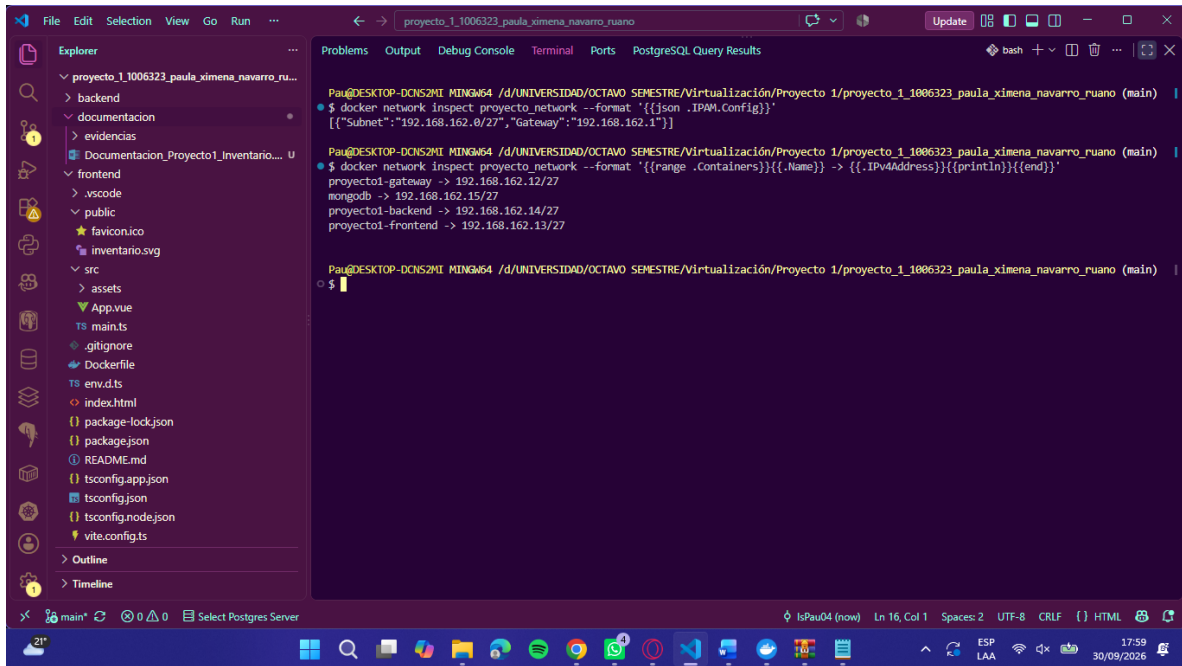


The screenshot shows a VS Code editor with a file explorer on the left and a terminal on the right. The file explorer shows the same project structure as the previous screenshot. The terminal shows the command 'docker inspect mongodb --format '{{range .Mounts}}{{.Type}} {{.Name}} -> {{.Destination}}{{println}}{{end}}'' being executed, which returns the volume information for the 'mongodb' container, showing it is using the 'proyecto1_mongo_data' volume mounted to '/data/db'.

```
Pau@DESKTOP-DCNS2HI MINGW64 /d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano (main)
$ docker inspect mongodb --format '{{range .Mounts}}{{.Type}} {{.Name}} -> {{.Destination}}{{println}}{{end}}'
volume proyecto1_mongo_data -> /data/db
volume f0364247b844049507e5747ed0f669c024870f53903d422e9a3e42d932a23faa -> /data/configdb

Pau@DESKTOP-DCNS2HI MINGW64 /d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano (main)
$
```

Verificación del montaje del volumen persistente en el directorio /data/db de MongoDB.



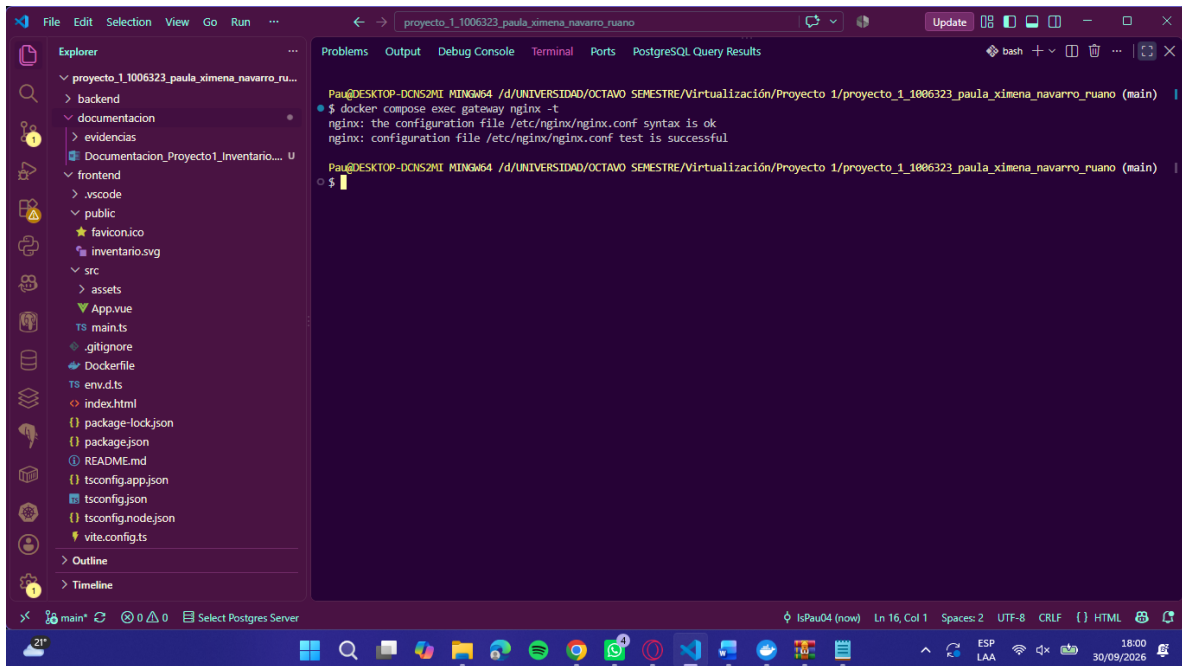
The screenshot shows a VS Code window with a terminal. The Explorer sidebar on the left shows a project structure for 'proyecto_1_1006323_paula_ximena_navarro_ruano'. The terminal has three command sessions:

```
Pau@DESKTOP-DCNS2MI MINGW64 /d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano (main) |
$ docker network inspect proyecto_network --format '{{.IPAM.Config}}'
[{"Subnet": "192.168.162.0/27", "Gateway": "192.168.162.1"}]

Pau@DESKTOP-DCNS2MI MINGW64 /d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano (main) |
$ docker network inspect proyecto_network --format '{{range .Containers}}{{.Name}} -> {{.IPv4Address}}{{println}}{{end}}'
proyecto1-gateway -> 192.168.162.12/27
mongodb -> 192.168.162.15/27
proyecto1-backend -> 192.168.162.14/27
proyecto1-frontend -> 192.168.162.13/27

Pau@DESKTOP-DCNS2MI MINGW64 /d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano (main) |
$
```

Verificación de la subred, puerta de enlace y direcciones IP de los contenedores en la red proyecto_network.



The screenshot shows a VS Code window with a terminal. The Explorer sidebar on the left shows the same project structure. The terminal has two command sessions:

```
Pau@DESKTOP-DCNS2MI MINGW64 /d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano (main) |
$ docker compose exec gateway nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful

Pau@DESKTOP-DCNS2MI MINGW64 /d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano (main) |
$
```

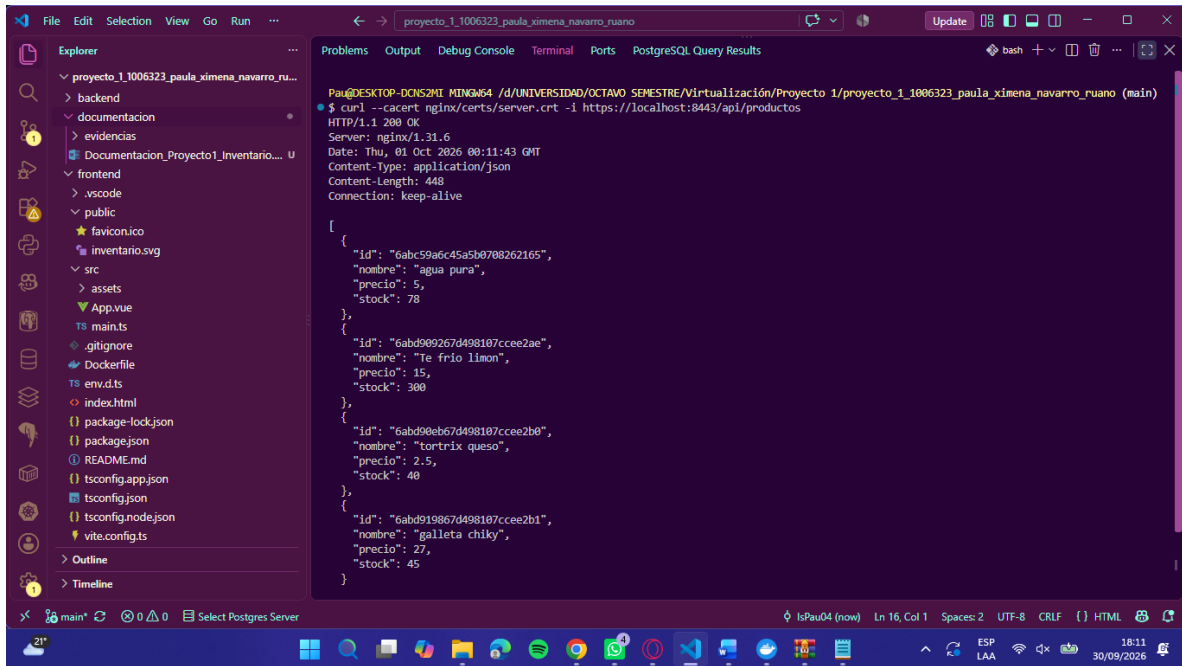
Validación de la configuración de Nginx en el contenedor gateway.

The screenshot shows a VS Code window with a terminal open. The terminal is running a command to generate a self-signed SSL certificate. The command is: `$ openssl x509 -in nginx/certs/server.crt -noout -subject -dates -ext subjectAltName`. The output shows the certificate details: `subject=CN = localhost`, `notBefore=Sep 30 00:55:23 2026 GMT`, `notAfter=Sep 30 00:55:23 2027 GMT`, `X509v3 Subject Alternative Name: DNS:localhost, IP Address:127.0.0.1`. The terminal is running on a Windows machine (MINGW64) in the directory `/d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano`.

Verificación del certificado SSL: identificación de localhost, vigencia y nombres alternativos.

The screenshot shows a VS Code window with a terminal open. The terminal is running a command to verify the SSL certificate. The command is: `$ curl -I http://localhost:8882`. The output shows the HTTP response: `HTTP/1.1 301 Moved Permanently`, `Server: nginx/1.31.6`, `Date: Thu, 01 Oct 2026 00:10:30 GMT`, `Content-Type: text/html`, `Content-Length: 169`, `Connection: keep-alive`, `Location: https://localhost:8443/`. The terminal is running on a Windows machine (MINGW64) in the directory `/d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano`.

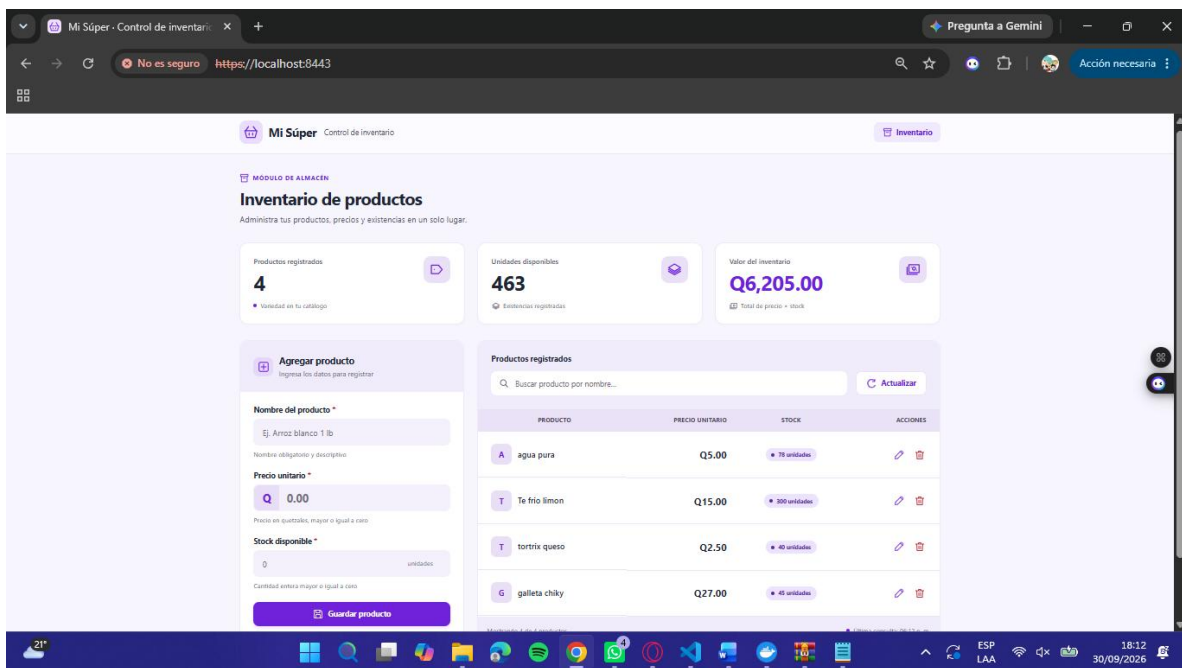
Verificación de la redirección automática de HTTP a HTTPS mediante Nginx.



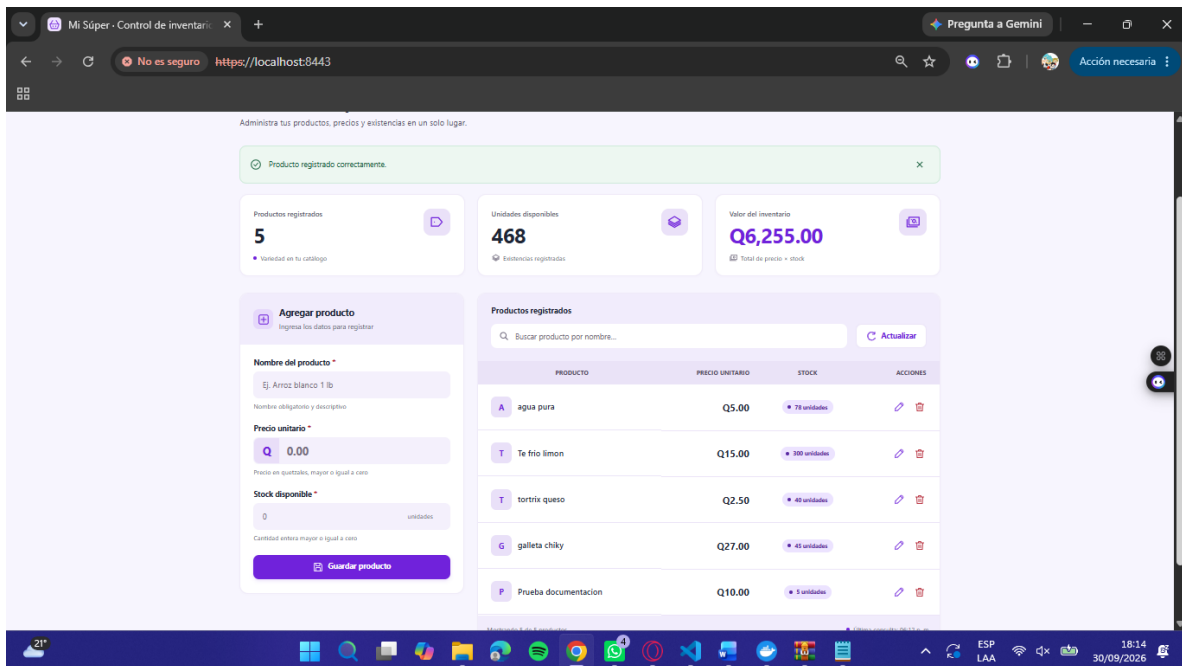
```
Pau@DESKTOP-DCNS2MT MINGW64 /d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano (main)
$ curl --cacert nginx/certs/server.crt -I https://localhost:8443/api/productos
HTTP/1.1 200 OK
Server: nginx/1.31.6
Date: Thu, 01 Oct 2026 00:11:43 GMT
Content-Type: application/json
Content-Length: 448
Connection: keep-alive

[
  {
    "id": "6abc59a6c45a5b8708262165",
    "nombre": "agua pura",
    "precio": 5,
    "stock": 78
  },
  {
    "id": "6abd909267d498107ccee2ae",
    "nombre": "Te frio limon",
    "precio": 15,
    "stock": 300
  },
  {
    "id": "6abd90eb67d498107ccee2b6",
    "nombre": "tortrix queso",
    "precio": 2.5,
    "stock": 40
  },
  {
    "id": "6abd919867d498107ccee2b1",
    "nombre": "galleta chiky",
    "precio": 27,
    "stock": 45
  }
]
```

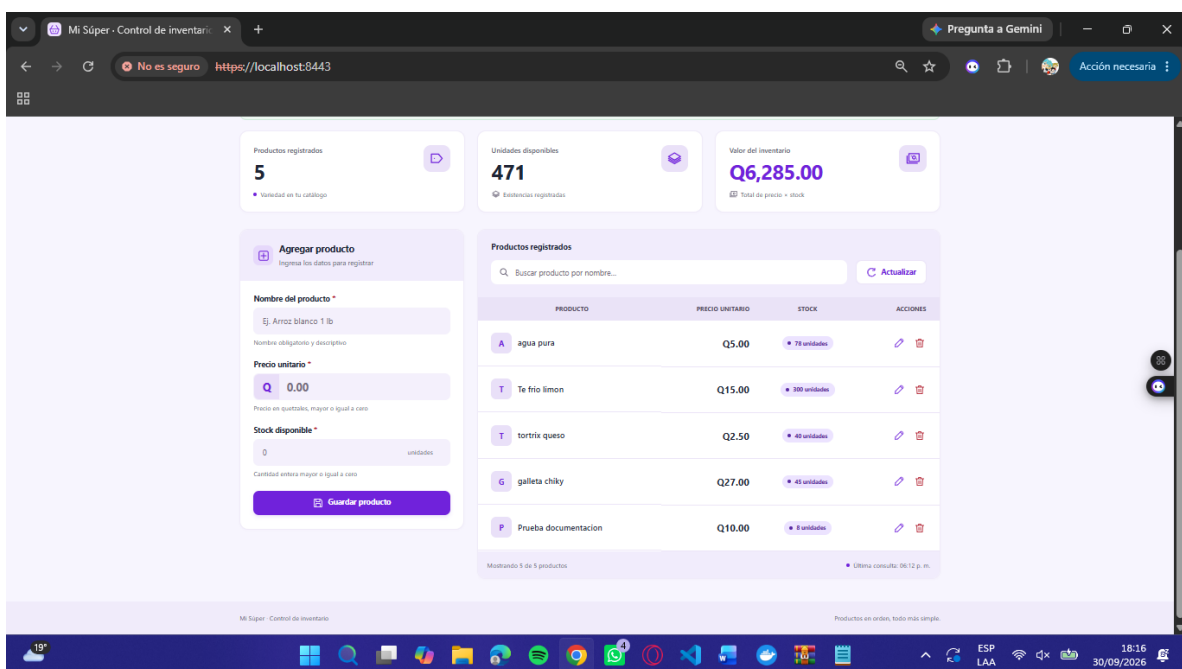
Consulta de productos mediante HTTPS con validación del certificado del servidor.



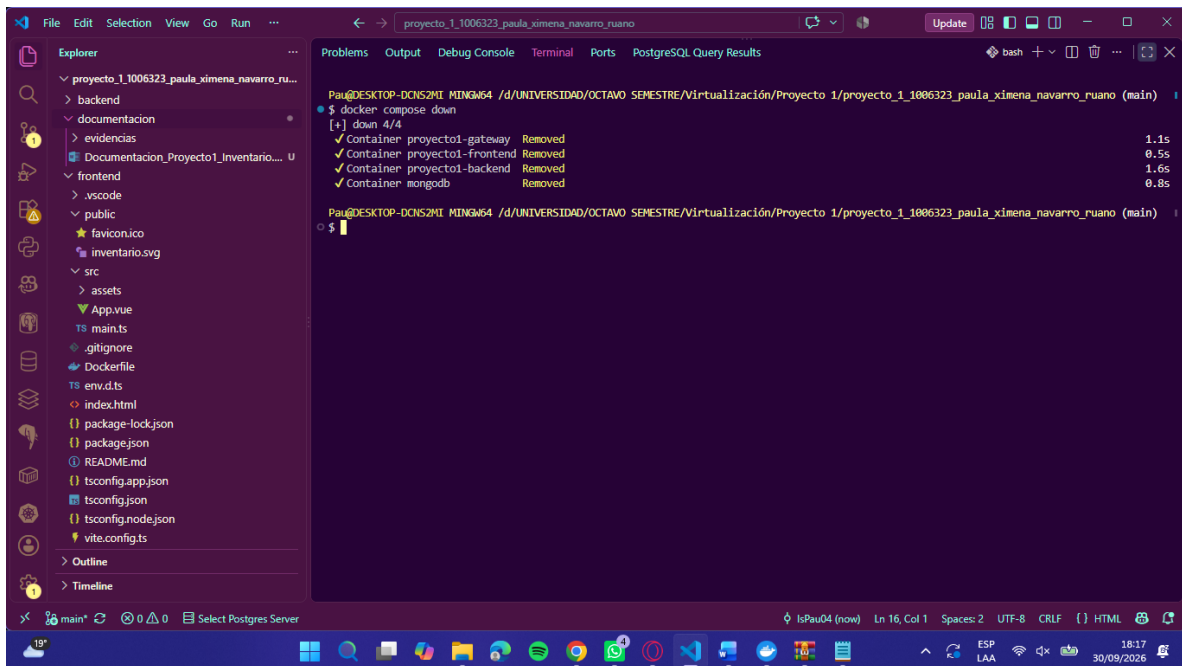
Acceso al sistema de inventario de supermercado mediante HTTPS.



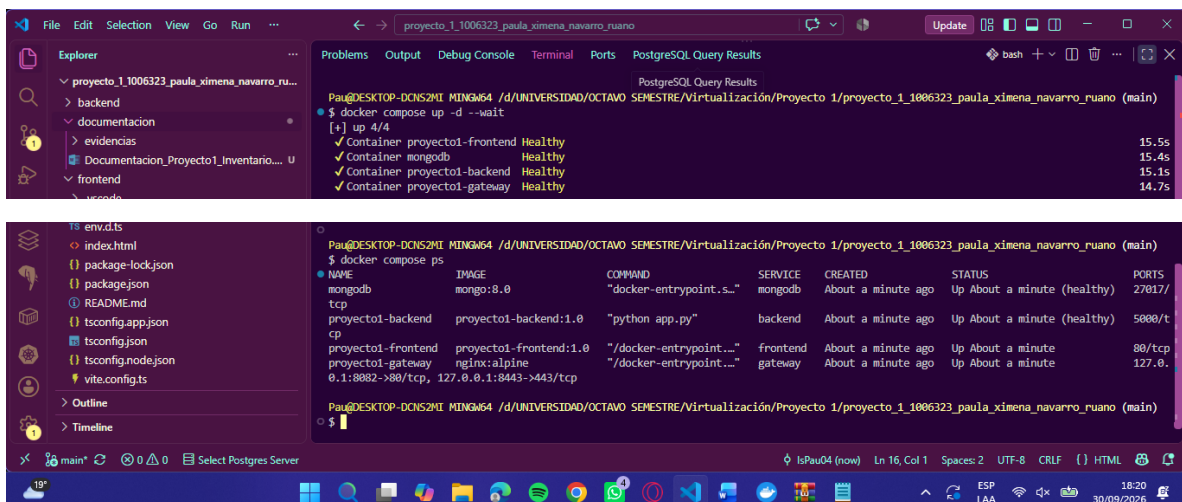
Registro de un nuevo producto desde el frontend del sistema de inventario.



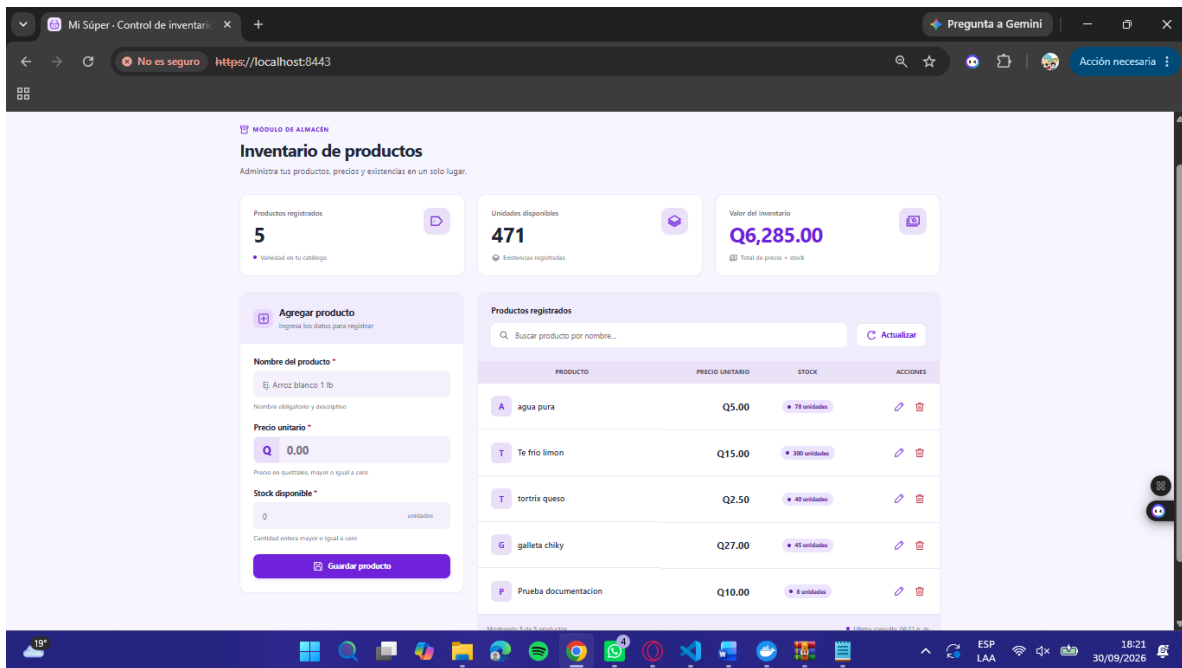
Actualización del stock de un producto desde el frontend del sistema de inventario.



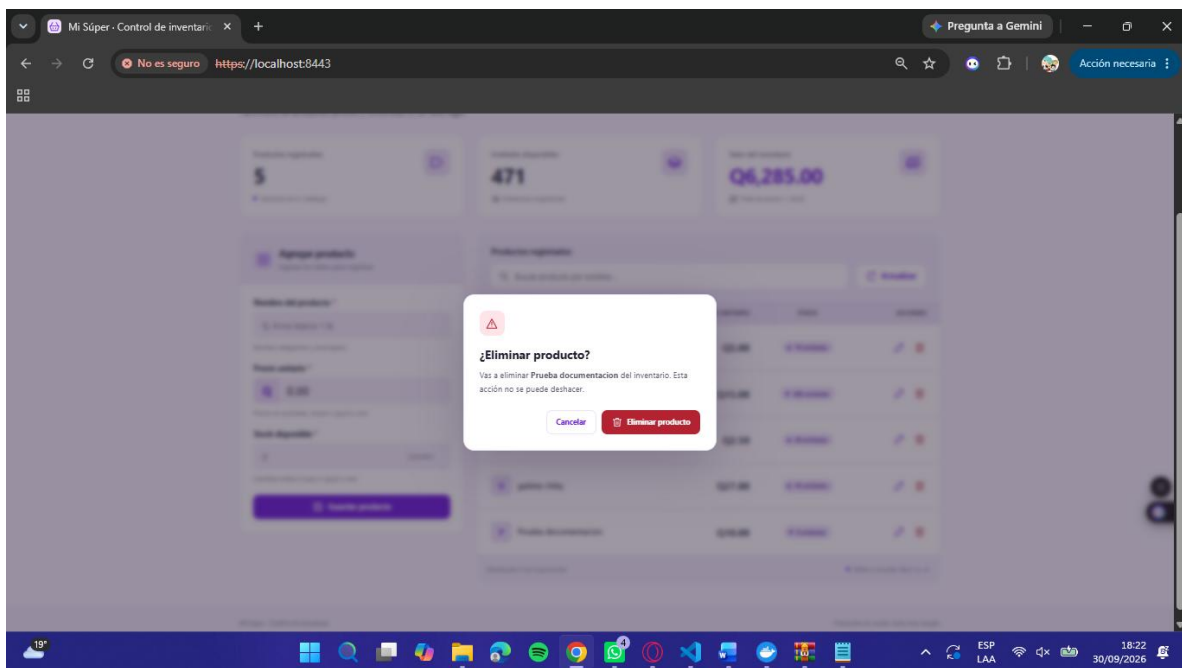
Detención y eliminación de los contenedores conservando el volumen persistente de MongoDB.

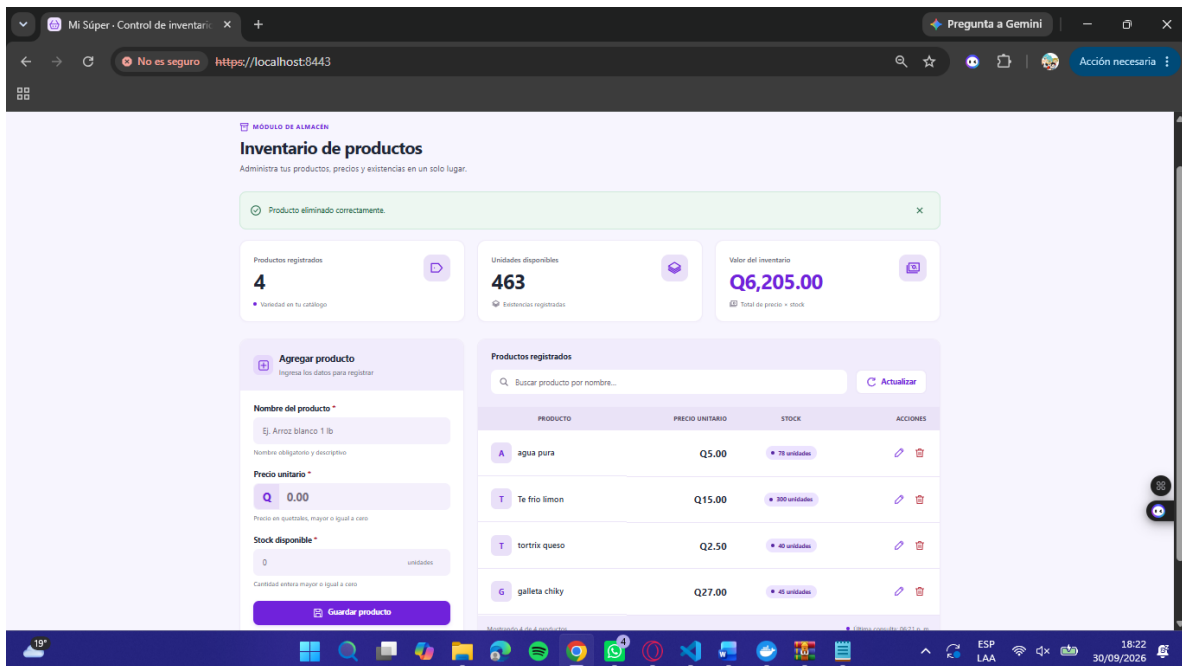


Recreación y puesta en marcha de los contenedores para verificar la persistencia de los datos.

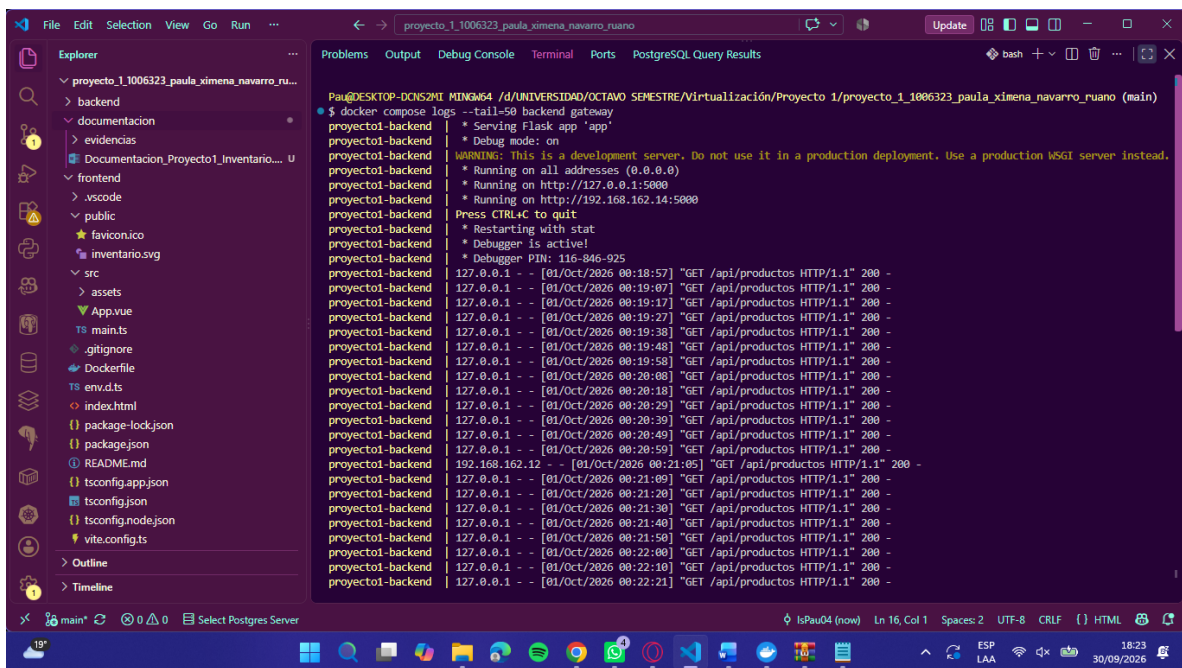


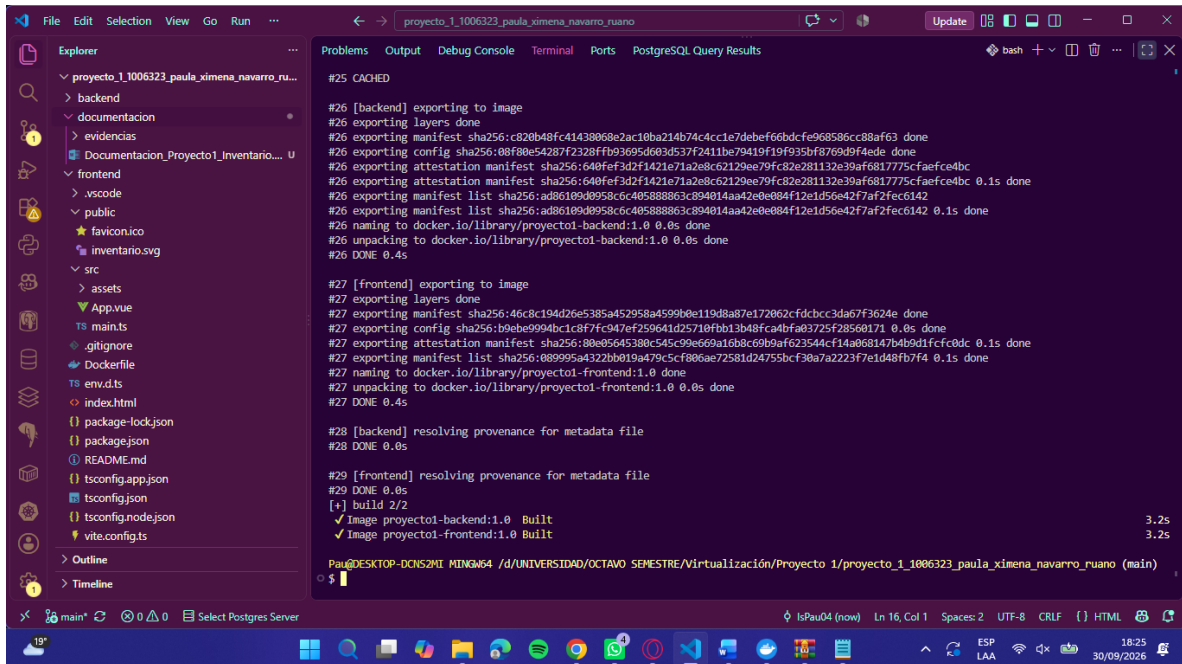
Verificación de la persistencia de los datos después de eliminar y recrear los contenedores.





Eliminación de un producto desde el frontend del sistema de inventario.





The screenshot shows the VS Code interface with a project named 'proyecto_1_1006323_paula_ximena_navarro_ruano'. The Explorer panel on the left shows the project structure, including folders for 'backend', 'documentation', 'evidencias', 'frontend', and 'public', along with various configuration files like 'App.vue', 'main.ts', 'tsconfig.json', and 'Dockerfile'. The Terminal panel on the right displays the output of a Docker build process. It shows the export of layers and manifests for both the backend and frontend images, with progress bars and timestamps. The build is completed successfully, with the final output showing the images built and pushed to Docker Hub.

```
#25 CACHED
#26 [backend] exporting to image
#26 exporting layers done
#26 exporting manifest sha256:c820b48fc41438068e2ac10ba214b74c4c1e7debe66bdcfe968586cc88af63 done
#26 exporting config sha256:80f80e54287f2328ff93695d603d537f2411be79419f19f935bf8769d9f4ede done
#26 exporting attestation manifest sha256:640fef3d2f1421e71a2e8c62129ee79fc82e281132e39af6817775cfaefce4bc done
#26 exporting manifest list sha256:ad86109d0958c6c405888863c894014aa42e0e084f12e1d56e42f7af2fec6142 0.1s done
#26 naming to docker.io/library/project01-backend:1.0 0.0s done
#26 unpacking to docker.io/library/project01-backend:1.0 0.0s done
#26 DONE 0.4s

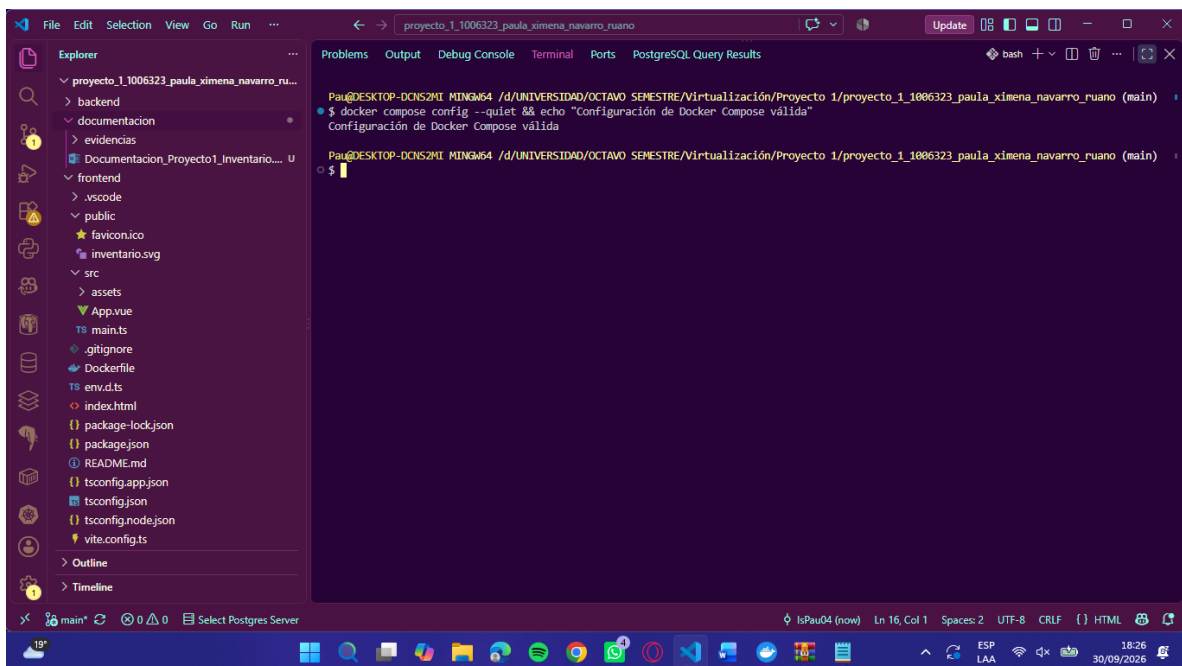
#27 [frontend] exporting to image
#27 exporting layers done
#27 exporting manifest sha256:46c8c194d26e5385a452958a4599b0e119d8a87e172862cfdcbcc3da67f3624e done
#27 exporting config sha256:b9e0e9994bc18f7fc947ef259641d25710fb113b48fca4bfa03725f28560171 0.0s done
#27 exporting attestation manifest sha256:80e05645380c545c9e669a16b8c69b8af623544cf14a068147b4b9d1fcfc0dc 0.1s done
#27 exporting manifest list sha256:889995a4322bb019a479c5cf806ae72581d24755bcf30a7a2223f7e1d48fb7f4 0.1s done
#27 naming to docker.io/library/project01-frontend:1.0 0.0s done
#27 unpacking to docker.io/library/project01-frontend:1.0 0.0s done
#27 DONE 0.4s

#28 [backend] resolving provenance for metadata file
#28 DONE 0.0s

#29 [frontend] resolving provenance for metadata file
#29 DONE 0.0s
[+] build 2/2
✓ Image project01-backend:1.0 Built 3.2s
✓ Image project01-frontend:1.0 Built 3.2s

Paul@DESKTOP-DCNS2MI MINGW64 /d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano (main)
$
```

Construcción de las imágenes Docker del backend y frontend mediante Docker Compose.

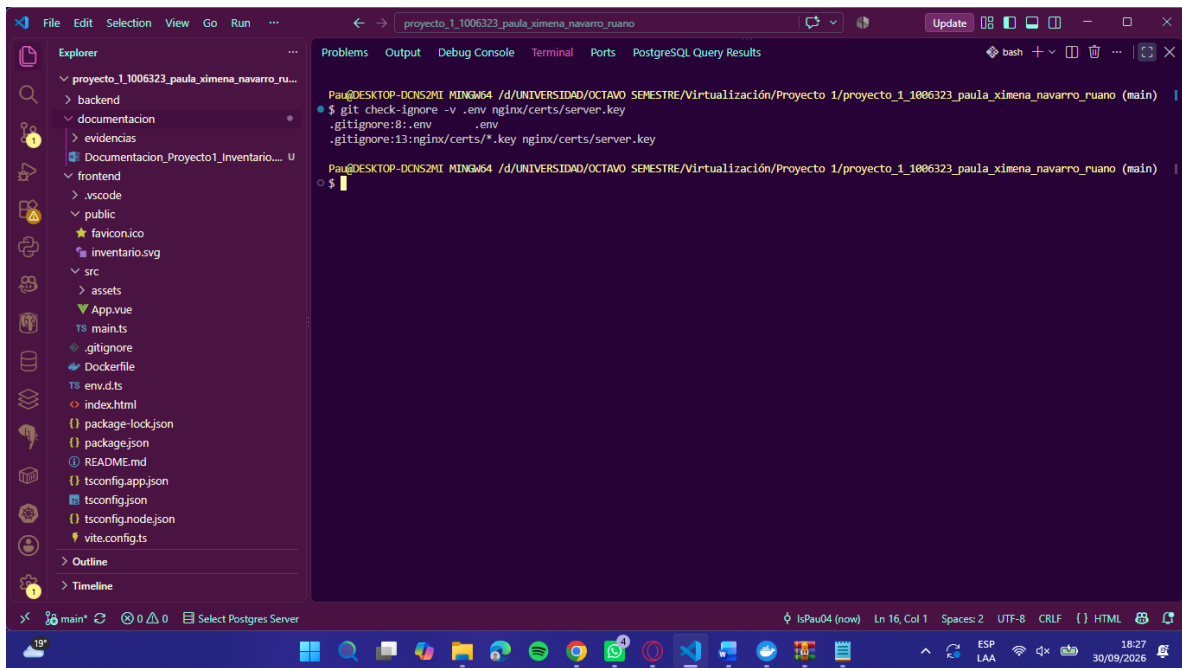


The screenshot shows the VS Code interface with the same project. The Terminal panel on the right displays the output of the 'docker compose config' command. The command is executed successfully, indicating that the Docker Compose configuration is valid. The output shows the configuration details for the 'project01' service, including the image name and the build context.

```
Paul@DESKTOP-DCNS2MI MINGW64 /d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano (main)
$ docker compose config --quiet && echo "Configuración de Docker Compose válida"
Configuración de Docker Compose válida

Paul@DESKTOP-DCNS2MI MINGW64 /d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano (main)
$
```

Validación del archivo docker-compose.yml y de las variables de entorno del proyecto.

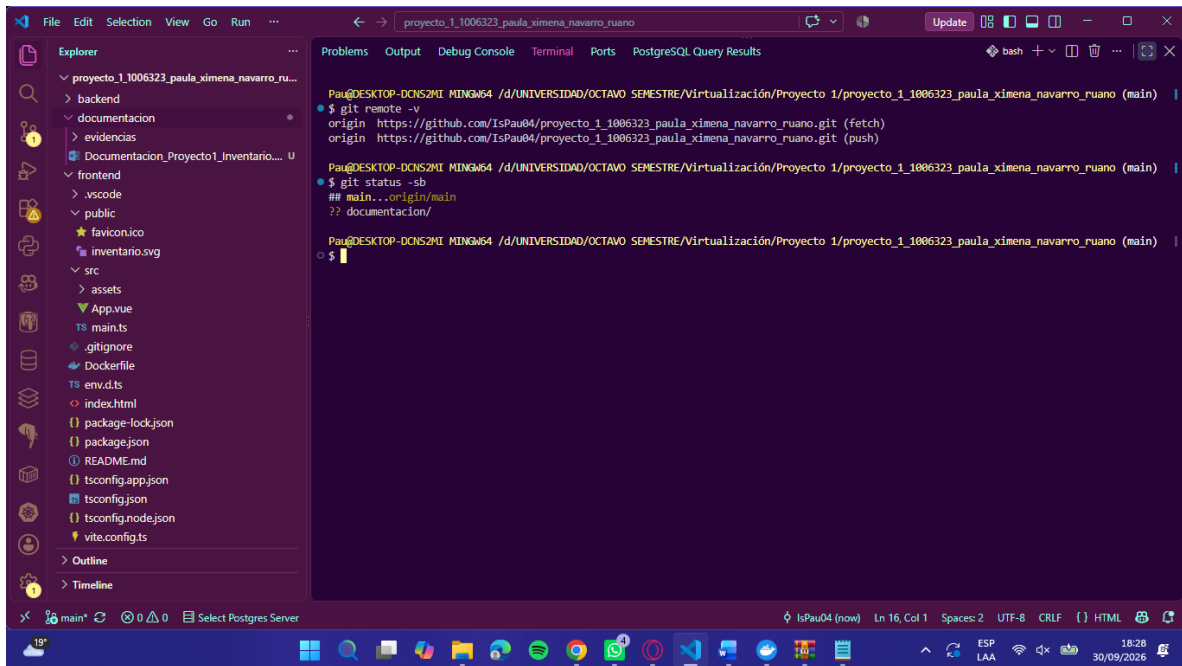


The screenshot shows the Visual Studio Code interface. On the left, the Explorer panel displays the file structure of the project 'proyecto_1_1006323_paula_ximena_navarro_ruano'. The 'documentation' folder is selected, showing files like 'evidencias', 'Documentacion_Proyecto1_Inventario...', 'public', 'faviconico', 'inventario.svg', 'src', 'assets', 'App.vue', 'main.ts', '.gitignore', 'Dockerfile', 'env.d.ts', 'index.html', 'package-lock.json', 'package.json', 'README.md', 'tsconfig.app.json', 'tsconfig.json', 'tsconfig.node.json', and 'vite.config.ts'. The main editor area shows the 'Terminal' tab with the following output:

```
Pau@DESKTOP-DCNS2MI MINGW64 /d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano (main) |
$ git check-ignore -v .env nginx/certs/server.key
.gitignore:8:.env
.gitignore:13:nginx/certs/*.key nginx/certs/server.key

Pau@DESKTOP-DCNS2MI MINGW64 /d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano (main) |
$
```

Verificación de la exclusión del archivo de variables de entorno y la llave privada del certificado mediante .gitignore.



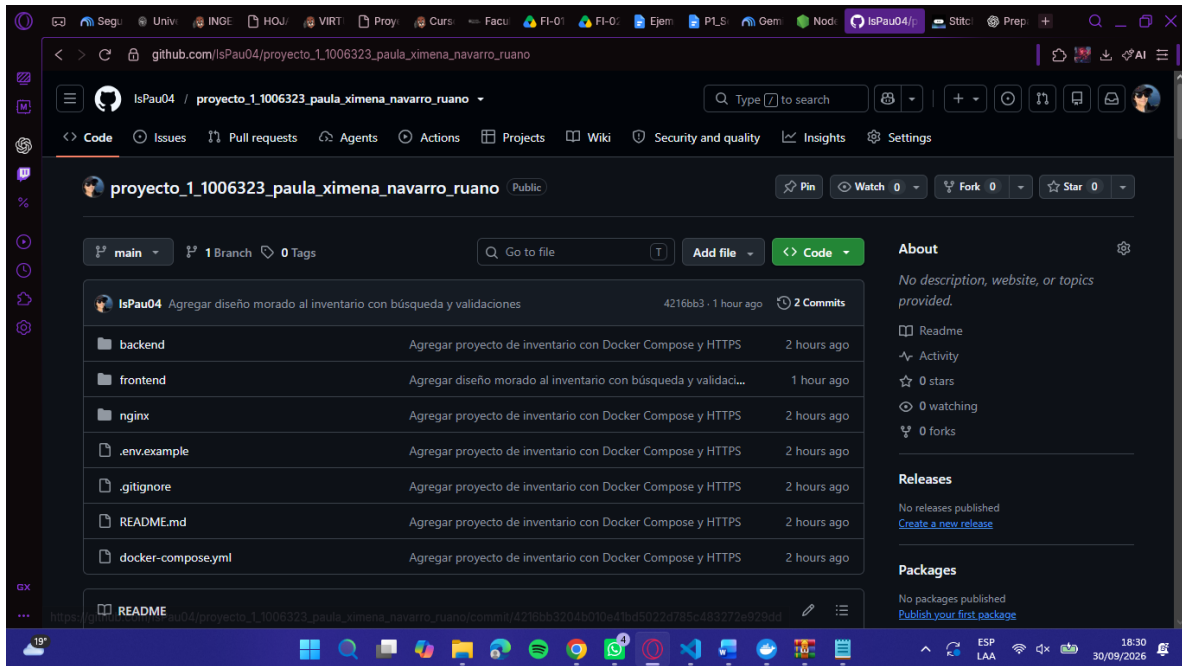
The screenshot shows the Visual Studio Code interface. The Explorer panel is the same as in the previous image. The main editor area shows the 'Terminal' tab with the following output:

```
Pau@DESKTOP-DCNS2MI MINGW64 /d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano (main) |
$ git remote -v
origin https://github.com/IsPau04/proyecto_1_1006323_paula_ximena_navarro_ruano.git (fetch)
origin https://github.com/IsPau04/proyecto_1_1006323_paula_ximena_navarro_ruano.git (push)

Pau@DESKTOP-DCNS2MI MINGW64 /d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano (main) |
$ git status -sb
## main...origin/main
?? documentacion/

Pau@DESKTOP-DCNS2MI MINGW64 /d/UNIVERSIDAD/OCTAVO SEMESTRE/Virtualización/Proyecto 1/proyecto_1_1006323_paula_ximena_navarro_ruano (main) |
$
```

Verificación del repositorio remoto en GitHub y del estado de la rama principal del proyecto.



Verificación del código fuente del proyecto publicado en GitHub.